Profile Depth and the Reviewing-to-Authoring Transfer Problem: Evidence from AI Expert Personas in Puzzle Hunt Design

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Abstract

Prior work established that reviewer profile depth is the dominant predictor of AI evaluation quality for puzzle hunt puzzles, with a non-monotonic gradient in which design philosophy captures most of the value of a full operational profile. Whether the same depth gradient governs AI *authoring* — and whether reviewing expertise transfers to creative generation — is an open question, because reviewing applies a lens to finished material while authoring must generate material that does not yet exist.

We conducted a 7-condition authoring ablation (A0–A6) mirroring a prior reviewer ablation, extended with a 3-designer × 2-mode comparison in which Snyder, Katz, and Young each authored puzzles under both a full authoring profile (A5) and their own C8-tier reviewing profile repurposed as authoring guidance (A6). All 13 conditions used a fixed knowledge-extraction task (Age of Empires II, answer TOWER, five-clue first-letter acrostic) evaluated by a same-session C8 expert panel on a 7-dimension weighted rubric (/45).

Three findings emerge. First, the authoring quality gradient mirrors the reviewer ablation: improvement is real but non-monotonic, with domain world data alone adding negligible value over bare task specification (A0 = 21.7, A2 = 28.3), and the largest single jump occurring at principle introduction (A2 to A3, +2.7 points). Second, transfer quality depends on lens type: operational lenses (Katz, +1.6 points) invert cleanly into authoring-time constraints; construction lenses (Snyder, −0.7) are already generatively oriented and require no transposition; perceptual lenses (Young, +3.7) are format-dependent — when the output format cannot supply the required visual material, the reviewer-as-author exercise forces a productive structural substitution, yielding the dataset high of 38.0/45. Third, a format ceiling constrains all A-series puzzles to Riven Standard ≤ 3: profile depth raises the quality floor within a fixed mechanism, but mechanism choice determines the ceiling.

The paper's primary contribution is a lens-type transfer model classifying reviewer profiles by generativity — construction, operational, perceptual — that predicts whether repurposing a reviewing profile for authoring will improve, preserve, or require reformulation. Empirical confirmation of the symmetric gradient establishes that the structural properties governing profile depth are not evaluation-specific. Practical guidance is given for constructing authoring profiles from reviewing profiles by lens type.

ACM Reference Format:
[Author]. 2026. Profile Depth and the Reviewing-to-Authoring Transfer Problem: Evidence from AI Expert Personas in Puzzle Hunt Design. In . ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Prior work established that reviewer profile depth is the strongest predictor of evaluation quality in AI expert panel assessments of puzzle hunt puzzles [?]. A bare reviewer persona — "you are a puzzle designer evaluating this puzzle" — produces serviceable but generic critiques; a fully elaborated profile incorporating domain knowledge, design philosophy, and a named lens vocabulary produces critiques that identify structural failures the bare condition misses entirely. The gradient is non-monotonic: domain expertise without a design framework adds negligible quality over bare baseline, while a philosophical worldview alone captures most of the value of the full profile. These findings motivate a natural question with a non-obvious answer: does the same depth gradient hold when the profile guides *creation* rather than critique?

The question is non-trivial because reviewing and authoring are cognitively distinct activities. A reviewer applies a lens to finished material; the lens is terminal, in the sense that it requires completed artifacts to have anything to evaluate. An author must generate material that does not yet exist; the profile must be generative, producing constraints that can be satisfied during construction rather than checks that can only be applied after the fact. A reviewer profile that asks "does the visual hierarchy resolve cleanly?" presupposes a layout to inspect; the same question offered as an authoring instruction is incoherent when the format is a text acrostic. The transfer from reviewing expertise to authoring expertise is therefore not guaranteed — and, we show, depends critically on the *type* of lens the reviewer applies.

We conducted a 7-condition authoring ablation (A0 through A6) that mirrors the reviewer ablation of Paper I exactly: A0 provides only a bare authoring persona, A1 adds task specification, A2 adds domain world data, A3 adds design principles, A4 adds an authoring philosophy, A5 adds a full authoring profile, and A6 repurposes a C8-tier reviewing profile as an authoring guide. All seven conditions produced puzzles evaluated by the same three-reviewer C8 panel (Katz, Snyder, Young) on a weighted 45-point rubric. We then extended the study with a 3-designer × 2-mode comparison: Snyder, Katz, and Young each authored puzzles in both their full

Conference'17, Washington, DC, USA

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

authoring profile (A5) and their own reviewing profile repurposed for authoring (A6), yielding six additional evaluated puzzles. The evaluation session was run with all 13 puzzles read before scoring to eliminate calibration drift.

Four findings emerged. First, the authoring quality gradient mirrors the reviewing quality gradient in shape: improvement is real but non-monotonic, with the largest single jump occurring at the principle-introduction boundary (A2 to A3, +2.7 points) and a plateau-and-retreat at the top of the Snyder ablation group (A5 at 32.0/45, A6 at 31.3/45). This symmetric gradient confirms that the mechanisms governing profile depth are not domain-specific to evaluation: the same structural properties of a profile — framework versus content, worldview versus checklist — determine quality in both directions. Second, A2 demonstrates an inversion that has practical implications: domain expertise without a design framework (28.3/45) is nearly indistinguishable from bare task specification (27.7/45). A data-rich profile that lacks generative structure produces technically accurate but experientially flat output — the clues read as database rows rather than puzzle constructions. Third, authoring worldview alone (A4, 30.7/45) is within 1.3 points of the full authoring profile (A5, 32.0/45), a gap smaller than any other adjacent condition pair except A0–A1. The full profile's marginal contribution over worldview is verification rigor — elimination of competing candidates, uniqueness checking — rather than the elevation of mechanism quality, confirming that framework matters more than completeness. Fourth, and most distinctively, the A5-to-A6 transfer result depends entirely on lens type: Katz's operational lens (named testable operations that invert cleanly to authoring constraints) improved by +1.6 points in A6 mode; Young's perceptual lens (checks requiring finished visual or experiential material) improved by +3.7 points — not because perceptual checks transferred, but because Young's reviewer-as-author exercise forced a novel solution within the format's constraints, replacing visual grammar with second-person narrative grammar; and Snyder's construction lens changed by only −0.7 points because his reviewing and authoring profiles are already functionally equivalent.

The lens-type result is the paper's primary theoretical contribution. We propose a three-category transfer model: *construction lenses* (Snyder) are already generative — the reviewing and authoring profiles collapse into one another; *operational lenses* (Katz) invert cleanly — each named test has a corresponding authoring-time constraint that can be applied before the puzzle exists; *perceptual lenses* (Young) are format-dependent — they succeed when the output format can supply the required material, and produce either failure or forced workarounds when it cannot. The model predicts which reviewer profiles can be safely repurposed for authoring tasks without redesign, which require inversion, and which require replacement.

The testbed is the same used in Paper I: puzzle hunt puzzles in the Age of Empires II domain, with fixed answer word (TOWER), fixed mechanism (five-clue first-letter acrostic), and locked world data. The controlled format enables clean ablation comparisons while exposing a format ceiling: all A-series puzzles cap at Riven Standard ≤ 3 because the first-letter acrostic mechanism is imported from puzzle hunt tradition rather than native to the AoE2 domain. This ceiling is itself informative — it establishes that profile depth raises the floor on execution quality within a fixed mechanism, while mechanism choice determines the ceiling, a distinction with direct implications for how authoring profiles should be deployed in practice.

The remainder of this paper is organized as follows. Section 2 situates the work relative to prior research on persona prompting, reviewer profile effects, and the authoring/reviewing distinction in human creativity research. Section 3 describes the authoring ablation design, the designer comparison methodology, and the evaluation rubric. Section 4 reports quantitative results across all 13 evaluated puzzles and introduces the six-framework taxonomy surfaced by richer profiles. Section 5 interprets the symmetric gradient, the lens-type transfer model, and the format ceiling. Section 6 concludes with practical recommendations for constructing authoring profiles from reviewing profiles, and identifies mechanism novelty as the open problem motivating future work.

2 Related Work

2.1 AI Reviewer Profiles and Profile Depth Effects

The use of large language models to simulate peer review has grown substantially as a strategy for scaling evaluation capacity in both academic and creative domains. Early work in this tradition optimizes for numeric accuracy relative to human judgments. Shah and Wein [?] investigate whether LLMs can replicate reviewer decisions on academic manuscripts, finding acceptable agreement on surface-level criteria but systematic degradation on assessments of novelty and contribution depth. Liu et al. [?] demonstrate that domain-targeted prompting improves the specificity of generated reviews, while the broader "LLM-as-judge" paradigm [?] establishes that LLM evaluators can approximate human pairwise preferences under controlled conditions. Berry et al. [?] extend this line to long-document structured review, showing that staged prompting improves review coherence but does not eliminate tendency toward generic praise in the absence of explicit evaluative schema.

Across this body of work, the target quantity is numeric proximity to human judgments, and persona is treated as an enabling mechanism rather than an independent variable. Profile richness is incidental rather than designed: reviewers are prompted with role descriptions sufficient to orient the model to the evaluation context, but the content and depth of those descriptions is not systematically varied. This leaves unexplored the question of whether profile specificity produces qualitatively different evaluative frameworks, not merely better-calibrated scores.

Our companion paper [?] addresses this gap directly in the evaluation setting, establishing that reviewer profile depth produces a non-monotonic quality gradient: a sparse generic profile outperforms the bare baseline, but domain-specific profiles with named analytical frameworks produce evaluative language that does not appear at lower depth levels. Critically, that study finds that the C2 condition — domain knowledge provided without evaluative framework — scores below the C1 condition (task specification only), a result we term the domain-inversion effect. The present paper asks whether the same gradient structure obtains when the same profiles are applied to authoring rather than reviewing, and whether the C2 inversion replicates in the authoring direction.

2.2 Persona Prompting for Creative Generation

The application of persona prompting to creative tasks has been studied from several angles. Zhang et al. [?] establish that explicit persona descriptions improve coherence and self-consistency in dialogue generation, confirming that LLMs respond to identity framing as more than surface decoration. Park et al. [?] demonstrate that memory-augmented persona representations support persistent, coherent behavior across multi-turn generative agent simulations, suggesting that the benefits of persona extend beyond single-turn generation and depend on the continuity of identity context across the interaction. Dang et al. [?] apply persona conditioning to creative writing assistance, finding that distinct persona instantiations produce meaningfully differentiated stylistic outputs from the same underlying model, with implications for diversity in AI-assisted co-creative workflows.

The central assumption of this literature is that professional identity functions as a holistic unit: the researcher is either instantiated or not, and instantiation activates the model's relevant prior knowledge as a package. Quality variation within a persona is not measured, and the constituent components of professional identity — domain knowledge, evaluative frameworks, philosophical disposition, characteristic cognitive moves — are treated as a single bundle that is present or absent. Whether a richer, more internally differentiated persona description produces qualitatively better outputs than a thinner one is not addressed.

A second unexamined assumption is that reviewing expertise and authoring expertise are interchangeable along the persona axis. When a profile describes how an expert evaluates creative work, it is not self-evident that the same profile will serve as a useful generative guide. A reviewer brings a terminal orientation to finished artifacts; an author requires a generative orientation to work-in-progress. Whether reviewing expertise is generative or terminal — whether it transfers cleanly to authoring, transfers partially, or introduces friction — depends on properties of the profile's cognitive lens that prior work has not isolated or studied.

2.3 The Authoring/Reviewing Distinction in Creativity Research

Human creativity research provides a theoretical foundation for expecting the authoring/reviewing distinction to matter in practice. Csikszentmihalyi's systems model of creativity [?] distinguishes the creator (who generates novel variations), the domain (the symbol system in which work is expressed), and the field (the social group of experts who select which variations are preserved). Creator and field operate with structurally different knowledge: the creator requires procedural knowledge of how to produce within the domain's constraints, while the field applies declarative knowledge of what the domain values and why. A reviewer is, in this model, a field actor; an author is, at minimum, a domain actor whose procedural fluency is the primary requirement.

Weisberg's analysis of creative expertise [?] reinforces this distinction at the cognitive level. Expert creative performance is not reducible to knowing what good work looks like; it requires internalized, practiced routines for generating candidate work and for diagnosing and repairing failures during generation. This is what Polanyi [?] distinguishes as knowing-how from knowing-that: the tacit procedural knowledge that underlies skilled practice cannot be fully recovered from declarative statements about quality standards, however detailed those statements are.

The authoring/reviewing distinction maps directly onto this. A reviewing profile operationalizes knowing-that: it articulates, often with considerable precision, what properties a finished puzzle should have. An authoring profile, if it is to function generatively, must encode something closer to knowing-how: the procedures, heuristics, characteristic moves, and generative impulses that a skilled constructor brings to a blank page. Profile depth may improve performance in both the authoring and reviewing directions, but the profile's content — specifically, whether its evaluative questions are *generative* (invertible to authoring constraints) or *terminal* (applicable only to finished artifacts) — should determine how much reviewing expertise survives the transfer. We term this property the lens type of a profile, and we make it the primary explanatory variable in our designer comparison study.

2.4 Puzzle Hunt Design as Creative Testbed

Puzzle hunt design is a tractable testbed for empirical work on creative generation for the same reasons it is tractable for evaluation: each puzzle has a single verifiable extraction answer, so correctness can be confirmed independently of quality; the expert community has produced a documented body of evaluative philosophy; and domain data can be precisely controlled through world files that fix the available material before authoring begins.

Several properties make this testbed especially useful for studying profile depth effects. The domain is narrow enough to support precise expertise claims — a reviewer with a stated specialization in puzzle construction either applies domain-appropriate evaluative criteria or does not, and this can be observed in the review text — but rich enough that shallow expertise produces recognizably different output from deep expertise. The AoE2 corpus used in Paper [?] provides a fixed domain world against which both authoring and evaluation outputs can be assessed for world-model consistency, a criterion that would be unavailable in domains without closed corpora. Practitioner frameworks from Snyder [?], Katz [?], and Young [?] provide the three designer profiles used in our comparison study, grounding the AI personas in authentic published perspectives rather than constructed approximations.

Taken together, prior work establishes that AI reviewer personas improve evaluation quality, that persona prompting affects creative generation, and that the human creativity literature provides principled reasons to expect the authoring/reviewing distinction to matter. What no prior study has examined is whether the quality gradient observed for reviewer profiles extends symmetrically to authoring profiles, whether the domain-inversion effect replicates in the generative direction, and — most distinctively — which properties of a reviewing profile determine whether it transfers productively to authoring or introduces lens-type friction. These are the questions the present paper addresses.

3 Methodology

3.1 Authoring Ablation Design

We constructed a seven-condition authoring ablation that mirrors, layer by layer, the reviewer ablation reported in Paper I [?]. Each condition adds exactly one element of authoring context to a fixed task; no condition removes context once introduced. The design isolates the marginal contribution of each context type while keeping the underlying task, domain, and evaluation rubric constant across all conditions.

A0 (Bare persona) provides only a minimal role declaration: *"You are a puzzle designer. Design a puzzle."* No domain, no mechanism, no format, no quality criteria. This condition establishes the floor — the output of role priming alone, without task specification or any domain knowledge.

A1 (Task specification) adds a complete task description: domain (Age of Empires II, Definitive Edition), mechanism (five-clue first-letter acrostic), fixed answer word (TOWER), and the scoring rubric against which the output will be evaluated. A1 is the minimum viable authoring prompt: a model that produces a puzzle satisfying A1 has received all logistically necessary information. The condition tests whether framework alone (task + rubric) can drive quality in the absence of domain grounding.

A2 (Domain world data) adds the full locked AoE2 world data: civilization roster with bonuses and unique unit/technology pairings, unit statistics, technology tree with costs and prerequisites, and map pool with terrain properties. No design guidance accompanies the data. A2 tests whether domain expertise in isolation — content richness without constructive framework — improves output quality over bare task specification, or whether the framework is the operative variable.

A3 (Design principles) adds all twenty design principles from the puzzle hunt toolkit `PRINCIPLES.md`, covering generative principles (Framing Is Structural, Reading Reward, Mechanism Owns the Domain) alongside craft principles governing clue fairness, extraction honesty, and solver trust. A3 represents the toolkit's contribution: domain-agnostic design wisdom that can be applied to any authoring context. The condition tests whether principled design guidance, without a specific designer's voice or construction sequence, is sufficient to cross the quality threshold.

A4 (Authoring philosophy) adds Thomas Snyder's generative worldview: his account of how he starts designing — backward from the target deduction, designing each constraint for what it closes off rather than what it states, structuring the solve path as a deliberate sequence rather than discovering it after clue-writing. The A4 profile contains no construction checklist; it captures worldview and starting orientation only. This condition isolates the contribution of a designer's philosophical frame, separable from the operational details of their construction practice.

A5 (Full authoring profile) adds Snyder's complete construction profile: his six-step design sequence (deduction identification, uniqueness verification, entry-point placement, solve-path tracing, decorative stripping, extraction verification), his signature moves (single solve path, designed entry point, load-balanced constraints, answer-key-first structural verification), his voice register (clinical and declarative, without explanation), and an explicit list of construction avoidances (decorative elements, multiple valid extraction paths, noise-based difficulty, incomplete verification, ambiguous clue phrasing, thematic decoration separable from mechanism). A5 is the richest generative condition and the primary comparison point for the designer comparison study described in Section 3.2.

A6 (Reviewer profile repurposed for authoring) substitutes Snyder's C8-tier reviewing profile — the same profile used as the most elaborated reviewer condition in Paper I — for the authoring profile of A5. The reviewing profile asks evaluation questions: does every step in the deduction sequence contribute load? Is the solve path unique? Are all constraints necessary and sufficient? A6 tests whether reviewing expertise transfers to authoring when the reviewing profile is the only generative context provided, and whether the profile's lens type mediates the quality of that transfer.

Table 1 summarizes all seven conditions, the context layers each provides, and the parallel reviewer condition in Paper I's ablation.

Table 1: Authoring ablation conditions (A0–A6) with cumulative context layers and parallel reviewer conditions from Paper I [?].

Cond.	Cumulative context	Content added
A0	Bare persona	Role declaration only
A1	+ Task specification	Domain, mechanism, answer word, scoring rubric
A2	+ World data	AoE2 civs, units, technologies, maps (no design guidance)
A3	+ Design principles	20 toolkit principles (`PRINCIPLES.md`)
A4	+ Authoring philosophy	Snyder generative worldview (backward design, deduction-first)
A5	+ Full authoring profile	Snyder construction sequence, signature moves, voice, avoidances
A6	Reviewer profile as author	Snyder C8 reviewing profile substituted for A5 content

3.2 Designer Comparison Design

We extended the ablation with a 3-designer × 2-mode comparison, yielding six additional authored puzzles. The three designers — Thomas Snyder, Dan Katz, and Dana Young — are the same domain experts whose C8-tier reviewing profiles were used in Paper I. Selecting the same experts for both studies enables direct cross-study comparison and controls for domain expertise as a confound: any quality difference between modes reflects the profile type, not the underlying expertise level.

Each designer authored puzzles in two conditions. The **A5 mode** used a full authoring profile constructed specifically for this study. Authoring profiles are generatively oriented: they ask how the designer *starts* designing (worldview and entry point), document their signature construction moves, characterize their voice and register, and enumerate what they avoid by instinct. These profiles are distinct from reviewing profiles in orientation and question frame, even when the underlying expertise is identical.

The three A5 profiles differ substantially in starting orientation. Snyder starts with the target deduction; Katz starts with the answer word and works backward from the aha moment; Young starts with the world, asking what it feels like to be inside the domain as an expert before selecting a mechanism. This divergence in starting orientation is not incidental — it structures the different construction sequences that follow and is the primary reason authoring profiles cannot be derived from reviewing profiles by simple inversion.

The **A6 mode** used each designer's existing C8 reviewing profile from Paper I, presented to the generation model as authoring guidance without modification. This condition tests whether the reviewing profile, which asks evaluative rather than generative questions, transfers to authoring without redesign, and whether the answer depends on the reviewing lens the designer applies.

3.3 Profile Lens Taxonomy

The designer comparison findings are structured by three lens categories that emerged from analysis of the A5 and A6 profiles across all three designers.

Operational lenses (exemplified by Katz) consist of named testable operations: the short-circuit check (can the solver guess the answer after two clues?), the Computer Test (is this the kind of puzzle a program could generate without domain knowledge?), the backwards read (does each clue uniquely identify its intended item when read in extraction direction?), and the Blame-the-Player test (after solving, does the solver feel they should have known each answer?). Operational tests are applied to finished puzzles as evaluation checks, but each inverts cleanly into an authoring-time constraint. "Would a solver short-circuit after clue 2?" becomes "ensure that no subset of $k < n$ clues reveals the answer"; "Does this clue work in any domain?" becomes "every clue must require AoE2-specific knowledge to solve." The inversion is direct and lossless.

Construction-facing lenses (exemplified by Snyder) ask questions about deduction uniqueness and load-bearing constraint structure. In a reviewing context, these are applied as checks: is the solve path unique? Does removing any single clue allow the deduction to fail? In an authoring context, these are already generative: "build each constraint so that removing it allows the deduction to fail" is an authoring instruction, not a post-hoc check. For construction lenses, the distinction between reviewing and authoring profiles collapses. The A5 and A6 profiles are functionally equivalent, which predicts (and is confirmed by) the near-zero score difference between Snyder's two modes.

Perceptual lenses (exemplified by Young) ask questions about visual grammar consistency, layout hierarchy resolution, and the visual landing of the final answer. These checks require finished visual or experiential material to apply: "does the layout hierarchy guide the eye to the entry point?" presupposes a layout. In a text-only format — a printed or plaintext puzzle hunt feeder with no graphic design layer — the material the perceptual lens requires does not exist. Perceptual checks cannot be applied during construction in this format, and they cannot be meaningfully inverted into authoring constraints when the format cannot supply the medium they require. This makes perceptual lenses format-dependent in a way that operational and construction lenses are not.

3.4 Authoring Task

All 13 conditions (A0–A6 plus 6 designer comparison conditions) used a single fixed authoring task. The domain was Age of Empires II (Definitive Edition), with a locked world data file containing the full civilization roster, unit statistics, technology tree, and map pool as used in Paper I. Locking the world data ensures that any quality difference between conditions reflects profile context, not variation in how models retrieve or weigh domain knowledge.

The mechanism was knowledge extraction: five clues, each identifying one AoE2 item (civilization, unit, technology, or map), where the first letter of each identified item spells the answer word. The answer word was TOWER, fixed across all 13 conditions. The output format was a complete puzzle hunt feeder puzzle as the solver would encounter it — problem statement, five clues, extraction instruction — plus a solution key identifying each intended AoE2 item and tracing each extraction letter.

3.5 Evaluation Rubric

Puzzles were evaluated on a seven-dimension weighted rubric totaling 45 points. Each dimension is scored 1–5 by the reviewer panel; Elegance and Reading Reward are double-weighted (each contributing up to 10 points) because both are the strongest tier discriminators in our calibration data from Paper I [**?**]. The remaining five dimensions are single-weighted (each contributing up to 5 points).

Clarity (1–5) measures whether a solver can determine what action the puzzle requires from the instructions alone, without ambiguity about the extraction mechanism or clue format.

Solvability (1–5) measures whether all five clues are resolvable by a solver with AoE2 domain knowledge, where a score of 5 requires that each clue uniquely identifies one AoE2 item and that no other item satisfies the clue description with the same initial letter.

Elegance (×2, 2–10) measures construction economy: a score of 5 (10 after weighting) requires that every clue phrase does double duty, constraining both the item's category and its specific identity, with no decorative language present and no clue phrase redundant with another.

Reading Reward (×2, 2–10) measures the quality of the solving experience — whether the puzzle produces the deductive satisfaction characteristic of well-constructed puzzle hunt feeders, rather than the experience of a domain knowledge quiz. High scores require that each clue lands as recognition ("of course") rather than lookup or guesswork.

Fun (1–5) captures overall solver enjoyment as rated by the panel using their own experience as the reference frame.

Confirmation (1–5) measures whether the answer word TOWER arrives as a satisfying conclusion — whether, at the moment of extraction, the answer reframes what the solver just did rather than appearing arbitrary.

Riven Standard (1–5) measures domain integration depth: a score of 1 indicates that the domain theme is decorative and any content domain could be substituted without changing the puzzle's mechanism; a score of 5 indicates that the mechanism is inseparable from the specific domain, and the puzzle could not exist outside of AoE2. The name references Riven and Myst as canonical examples of world-mechanism integration in game design. All A-series conditions using the first-letter acrostic mechanism are expected to cap below 4 on this dimension, since the acrostic is imported from puzzle hunt tradition rather than derived from AoE2's internal logic.

The rubric is a proxy for creative quality, not a direct measure. The double-weighting of Elegance and Reading Reward reflects their alignment with established computational creativity criteria: Elegance maps to Boden's *combinational creativity* criterion [?] — novel but valued recombination of existing elements, where economy of expression signals that the combination is doing genuine work — and Reading Reward maps to the SPECS framework's *domain-competent* criterion [?], which requires that outputs demonstrate knowledge of the domain rather than merely invoking its surface vocabulary. The Riven Standard dimension operationalizes Boden's *exploratory creativity* criterion: working within an established conceptual space (here, AoE2's unit and technology relationships) rather than importing structure from outside it. The rubric has face validity established against 15 real puzzle hunt puzzles spanning three quality tiers in Paper I's Experiment 4 [?], where the same seven dimensions separated tier populations with no overlapping score distributions.

The pass threshold is a panel average of 33/45 (73% of maximum; this ratio matches the 22/30 = 0.733 threshold used in Paper I's reviewer ablation [?], which was calibrated against the quality level at which puzzles in well-regarded MIT Mystery Hunt and Microsoft Puzzlehunt rounds have shipped). All 13 puzzles were evaluated by the same three-reviewer C8 panel (Katz, Snyder, Young, each presented with trimmed credentials and three injected design principles to fix the evaluation context). All 13 puzzles were read by the panel before any scores were assigned, ensuring calibration consistency across conditions and eliminating the session-order drift that would otherwise make cross-condition score comparisons unreliable.

4 Results

4.1 Authoring Profile Ablation (A0–A6)

Table 2 reports panel-averaged scores across the seven authoring conditions. The gradient confirms the paper's first claim: authoring profile depth produces measurable quality improvement in the same non-monotonic pattern documented for reviewing profile depth in Paper I.

Table 2: Authoring ablation results. Scores are panel averages on the weighted 45-point rubric (Clarity ×1, Solvability ×1, Elegance ×2, Reading Reward ×2, Fun ×1, Confirmation ×1, Riven Standard ×1). Pass threshold: ≥ 33/45. Range = min–max across the three reviewers; differences within this spread should be interpreted with caution given the small panel size.

Condition	Context Added	DK	TS	DY	Avg/45	Pass	Range
A0	None (bare persona)	22	21	22	21.7	N	21–22
A1	Task specification	28	27	28	27.7	N	27–28
A2	Domain world data	28	29	28	28.3	N	28–29
A3	Design principles (all 20)	30	32	31	31.0	N	30–32
A4	Authoring philosophy	30	33	29	30.7	N	29–33
A5	Full authoring profile	33	35	28	32.0	N	28–35
A6	Reviewer profile as author	31	35	28	31.3	N	28–35

The overall arc spans 9.6 points from A0 (21.7) to A5 (32.0), a range large enough to be meaningful against a rubric with an interior pass threshold at 33. The four lowest conditions (A0–A3) form a consistent improvement sequence, with the largest single step occurring at the principle-introduction boundary: A2 to A3 adds 2.7 points (28.3 to 31.0), the same structural jump seen in Paper I's reviewer ablation when principles were introduced at C3. No condition in the A-series passes, however. The highest aggregate score is 32.0, one point below threshold. This is not a scoring artifact: all seven A-series puzzles score Riven Standard ≤ 3/5, and since the rubric double-weights Elegance and Reading Reward while treating Riven Standard as a single-weight multiplier, the format ceiling propagates through the entire top of the score distribution. The first-letter acrostic mechanism is imported from puzzle hunt tradition; it is not native to the Age of Empires II domain. No authoring profile, however fully elaborated, can substitute domain-native mechanism design when the rubric specifically rewards mechanism-domain integration.

The most diagnostically important result in the ablation is the near-identity of A1 and A2. A2 adds extensive domain world data — every clue carries factual citations to AoE2 source files — yet scores only 0.6 points above A1 (28.3 vs. 27.7; reviewer range for A2: 28–29, for A1: 27–28), a difference within the panel's typical inter-reviewer spread and therefore best treated as a directional signal rather than a precise measurement. The evaluation panel converges on a consistent explanation: the A2 puzzle, *Stone and Steel*, reads like a reference table with punctuation added. Its clues are factually unimpeachable; the Throwing Axeman's ranged-infantry paradox and the War Elephant's 600-hit-point ceiling are both verifiable and specific. But factual specificity is not the same as constructive tension. The clues do not make use of their facts to create an experience — they deliver the fact and stop. Snyder describes this as clues that read as "excerpts from a reference table rather than constructions"; Young notes that only one clue (War Elephant) carries a tactile surprise that causes a player to lean in. Domain expertise without a design framework passes the Computer Test for accuracy while failing the Computer Test for craft: a script can retrieve those facts, and a script retrieving them would produce the same clue texture. The gap from A2 to A3 — when design principles are introduced — is more than four times the gap from A1 to A2, confirming that the framework matters more than the content it organizes.

The other notable non-linearity is the A4–A5–A6 plateau. A4 (authoring philosophy alone, 30.7; reviewer range: 29–33) is within 1.3 points of A5 (full authoring profile, 32.0; reviewer range: 28–35) — the smallest gap between any adjacent condition pair after A1–A2, though the overlapping reviewer ranges at A4 and A5 indicate this difference should be read as a weak trend rather than a sharp boundary. The full profile's contribution over philosophy is primarily verification: A5's construction log documents systematic elimination of competing candidates for each letter position, exhaustive uniqueness checking, and instinct-avoidance accounting. This is quality control rather than quality generation. A designer who has internalized the worldview (*BREACH*: strip each clue to the single irreducible fact that names the item; test every phrase for load-bearing status; let the title do double work) is already generating at near-peak quality for the format. The verification layer protects that quality from construction errors but does not elevate the mechanism.

The retreat from A5 to A6 (32.0 to 31.3) is the expected consequence of repurposing a reviewing profile as an authoring guide when the reviewer's most distinctive instinct — the coherence-and-coupling standard — exposes a structural gap the format cannot close. The A6-Snyder puzzle, *Escalating Force*, is intellectually honest about this: its authoring notes document the gap between the acrostic mechanism and Snyder's domain-integration standard. That honesty is admirable and accurate, but it describes a limit rather than solving one. The reviewer-as-author transposition adds metacognitive overhead to the authoring process without adding solver-visible quality to the result.

The gradient shape mirrors the reviewer ablation gradient from Paper I closely enough to support the symmetric-gradient hypothesis (Figure 1). The same three structural features appear in both: an early-plateau region where domain data adds little over task specification, a principle-introduction jump, and a high-conditions plateau where additional profile depth produces diminishing returns. The mirror is not exact — the authoring plateau is lower in absolute terms because the format ceiling caps the A-series below the pass threshold — but the functional shape is preserved.

4.2 Designer Comparison (A5 vs. A6 $\times$ 3 Designers)

Table 3 reports scores for all six conditions in the designer comparison, alongside the A5-to-A6 transfer gap for each designer and the lens-type classification introduced in Section 3.3.

Table 3: Designer comparison results. A5 = full authoring profile; A6 = reviewer profile repurposed for authoring. Gap = A6 score minus A5 score (positive indicates improvement under A6 mode). Lens type: C = construction, O = operational, P = perceptual.

Designer	A5/45	Pass	A6/45	Pass	Gap	Lens
Snyder	32.0	N	31.3	N	−0.7	C
Katz	33.7	Y	35.3	Y	+1.6	O
Young	34.3	Y	38.0	Y	+3.7	P

The most striking result is Young's gap. Moving from A5 (34.3) to A6 (38.0) adds 3.7 points — more than twice Katz's gain and of opposite sign to Snyder's. *Holding the Line*, the A6-Young puzzle, is the highest-scoring puzzle in the entire evaluated set and receives the only perfect Riven Standard score (5.0 average) and the only perfect Elegance score (5.0 average) across the full 13-puzzle corpus. It achieves this by finding a format-compatible substitute for Young's most distinctive instinct. Young's reviewing lens centers on visual grammar, world-model consistency, and what the outline calls perceptual-shift — the moment when the answer produces a second reading that the domain knowledge alone makes available. In a text acrostic, visual grammar has no material to work with, and a pure visual-resolution check would fail immediately. The reviewer-as-author exercise forces a different question: what, within the format's constraints, can perform the function that visual grammar performs in a richer format? Young's answer is second-person present-tense narrative. "You are playing Age of Empires II. You queue this Castle Age research…" puts the solver inside a game decision rather than presenting trivia. The five items — Treadmill Crane, Onager, Watch Tower, Elite Skirmisher, Ring Archer Armor — are chosen not for their first letters but for their roles in a coherent defensive scenario; TOWER arrives as the name of the strategic role the solver has been defending throughout, which is a precise analog of visual resolution in the narrative register. The perceptual instinct transferred, but only because the reviewer-as-author exercise generated a format-specific workaround rather than importing the original check.

Katz's improvement is more direct. His operational lens — short-circuit check, Computer Test, variety constraint, backwards-read discipline — inverts cleanly into authoring constraints. Each named test has an authoring-time analog: the short-circuit check becomes a constraint on clue ordering (do not let the first three letters become unambiguously suggestive before the fourth), the Computer Test becomes a candidate-selection filter (eliminate any item whose defining property is retrievable by a script), and the backwards-read test becomes a verification pass applied after drafting. The A6-Katz puzzle, *WOLOLO: Five of a Kind*, shows all three constraints operating. The title is the most domain-saturated in the set: WOLOLO names the Monk conversion sound, which is an AoE2-specific piece of community knowledge rather than a game mechanic, establishing the world before the solver reads a clue. Clue 2's Onager friendly-fire detail is a piece of veteran community memory — a non-player retrieving the Onager's unit table would find range and damage values, not the friendly-fire note — passing the Computer Test by requiring social knowledge rather than data knowledge. The Snyder Computer Test, imported by Katz into his authoring pass, functions as a domain-integration filter: it eliminates technically correct clues that do not require the solver to have lived in the world. The 1.6-point gain from A5 to A6 is modest in absolute terms but significant as evidence that operational tests are generative — they add authoring value when repurposed from reviewing.

Snyder's near-zero gap confirms the construction-lens prediction. His reviewing profile asks whether each element is load-bearing, whether the solve path is singular, and whether the cross-clue dependencies form a coherent deduction chain. These are already authoring questions: they can be asked and answered before the puzzle exists in finished form. The A5 and A6 profiles generate functionally equivalent puzzles because they apply functionally equivalent constraints. The −0.7-point retreat in A6 mode reflects the metacognitive overhead noted in Section 4.1: the reviewer-as-author exercise causes the A6-Snyder puzzle to document its own limitations rather than resolve them, which is accurate self-assessment but not a design improvement. For a designer whose reviewing and authoring lenses are already unified, the reviewer-as-author transposition adds nothing and costs a small amount of authoring attention to the metacommentary.

Taken together, the three gaps confirm the lens-type transfer model. Construction lenses (Snyder, −0.7) require no transposition because they are already generative. Operational lenses (Katz, +1.6) transfer cleanly because named tests have authoring-time analogs. Perceptual lenses (Young, +3.7) are format-dependent: when the output format can supply the required material, the reviewing instinct is powerful and produces the highest scores in the corpus; when it cannot, the reviewer-as-author exercise must generate a format-specific substitute, which is cognitively demanding but, in Young's case, productive.

4.3 The Six Evaluative Frameworks

The quantitative gradient reported in Sections 4.1 and 4.2 is supported by a qualitative finding: richer reviewer profiles introduce new evaluative *language*, not merely more words about existing criteria. Paper I's C6 reviews — produced by the same three-reviewer panel under its richest profile condition — surfaced six evaluative frameworks that do not appear in C0–C5 reviews at any scoring level. These frameworks are not refinements of existing rubric dimensions; they name new constructs that the bare and intermediate profiles lack the vocabulary to express. Table 4 presents the taxonomy.

Table 4: Six evaluative frameworks surfaced in C6 reviews. None appear in C0–C5 reviews. Full empirical evidence — review text instances, frequency analysis, cross-condition comparison — is provided in the companion taxonomy paper [?].

Framework	Originating Profile	Definition
A-ha economy	Foggy Brume	The design and pacing of insight moments: not merely whether a puzzle has an a-ha, but whether the density and timing of a-ha moments match the puzzle's promised experience.
Load-bearing test	Thomas Snyder	Every element must earn its structural place; strip it and test whether the puzzle collapses. Elements that survive removal were decorative; elements that collapse the puzzle are load-bearing.
World-model consistency	Emily Short	The puzzle's internal logic must hold within the fiction it inhabits; a clue that violates the world's own rules breaks the contract with the solver even if the extraction is mechanically correct.
Social fabric	Jordan Weisman	A puzzle genuinely requiring collaboration is architecturally different from a puzzle that merely occurs in a team context; the social fabric test distinguishes puzzles that would degrade if solved alone.
Perceptual-shift	Scott Kim	The answer should produce a second reading of the puzzle's surface that the domain knowledge makes available and that a non-player cannot experience; the reward is not recognition but transformation.
Visual grammar	Dana Young	The layout should communicate structure through form; when text substitutes for visual design, the puzzle is describing its structure rather than enacting it.

The following exemplars from the C6 review corpus illustrate each framework operating as an analytical lens on a specific puzzle feature.

A-ha economy. Dan Katz on *Scicabulary* (MIT Mystery Hunt 2023):

> The gap between understanding the instruction and executing the first successful match is substantial. A solver who does not immediately grasp what "opposite halves" means will stall at the threshold. The canned hints address this, but that means the puzzle is partially relying on its hint structure to compensate for a presentation that could be clearer. Once you are in — once you've cracked two or three pairings — the mechanism is deeply satisfying.

The Clarity score (3/5) does not capture this diagnosis: the framework identifies a timing displacement between entry and payoff that the rubric dimension cannot resolve.

Load-bearing test. Thomas Snyder on *What's Next?* (Huntinality III 2023):

> The ten images represent a deliberate sample of the song's timeline, and the designer chose images where the "next item" yields letters that spell a coherent phrase. That sample selection is load-bearing design: not any ten entries would have worked, and these ten were chosen because they combine to spell a recognizable lyric.

The test here operates at the construction-architecture level, classifying the entry-selection decision itself as structural rather than arbitrary.

World-model consistency. Dana Young on *Information Relay* (Huntinality III 2023):

> The five hearts each have distinct voices, and those voices are mechanically load-bearing — the personality is the clue method. That is the mechanic being the theme in the purest sense I've seen in this batch. You spend the entire solve inhabiting a family dynamic where everyone is contributing to a shared task in their own broken way.

The framework diagnoses thematic enactment rather than thematic accuracy: the fiction and the mechanism are the same object.

Social fabric. Dan Katz on *Bridge Building* (MIT Mystery Hunt 2023):

> This two-stage structure is one of my favorite puzzle architectures because each stage rewards different solvers. A chemistry-blind team can get the Hashi; a chemistry team might backsolver from the polymer structure.

The framework distinguishes architectural diversity-of-entry from difficulty variation: distinct roles arise because logic and chemistry competencies are uncorrelated.

Perceptual shift. Dana Young on *Front and Center* (Huntinality III 2023):

> The answer RADAR is perfect: it names a technology for detecting things that are both ahead and behind you, which resonates with the temporal symmetry of palindromes. This is Dana's definition of an inevitable answer: it reframes everything before it.

The framework captures what a satisfaction rating cannot: the reorientation is available only to a solver who has worked through the palindrome mechanism.

Visual grammar. Dana Young on *Front and Center*:

> This puzzle resolves visually at every scale: each palindrome headline reads the same forward and backward, the extracted word RADAR reads the same forward and backward, and the masthead crease that bisects the T in SEMI TIMES is a visual hint at the center-letter extraction. The visual grammar is consistent throughout. The newspaper format is not decorative — it is the world of the puzzle.

The failure mode appears in Young's review of *H2No*: "the visual presentation of this puzzle is doing almost no work to guide you through the intended experience. The layout is a list" — a format-level diagnosis that survives all content-level corrections.

These frameworks are the qualitative evidence for a claim that the quantitative gradient alone cannot establish: that richer profiles change the *kind* of evaluation being performed, not just its thoroughness. A C0 reviewer can identify that a clue is unclear; a C6 reviewer can identify that the clue's clarity problem is structural — it describes the world-model relationship incorrectly, which cannot be fixed by rewording. The a-ha economy framework allows a reviewer to diagnose pacing failure as distinct from content failure; the load-bearing test distinguishes decoration from structure at the element level; visual grammar identifies format-level mismatches that survive all content-level improvements.

Within the present study, three of these frameworks appear directly in the A5/A6 designer evaluation notes. The load-bearing test is explicit in both Snyder's authoring log (A5-Snyder: "each phrase was tested for load-bearing status") and Katz's reviewer notes on A6-Katz ("every clue is forced"). Perceptual-shift appears in Young's A6 construction log as the explicit design goal that second-person narrative grammar is intended to satisfy: the answer TOWER should arrive as a transformation of the solver's experience of the preceding scenario, not as the conclusion of a letter-reading procedure. Visual grammar is the framework whose absence is most diagnostic: none of the text-acrostic puzzles can satisfy it, and the Riven Standard scores for the A-series (capped at 3/5 for all seven conditions) reflect precisely this gap between content-domain integration and format-domain integration. The framework does not appear in A-series reviewer notes as a positive finding; it appears in Young's A5 notes as an acknowledged absence, and its workaround in A6-Young is what produces the highest Riven Standard score in the corpus (5.0).

Table 5 reports framework presence counts across all 15 C6 reviews compared to the C0 bare baseline, confirming that all six frameworks are emergent from the enriched profiles rather than present in unprompted reviews.

Full empirical evidence for the six-framework taxonomy — review text instances, frequency analysis by profile condition, and cross-condition presence/absence comparison — is provided in the companion taxonomy paper [?]. Here we present the taxonomy as interpretive context for the lens-type patterns reported in Section 4.2: the three lens types (construction, operational, perceptual) are the authoring-transfer manifestations of the same underlying profile properties that generate the six evaluative frameworks on the reviewing side.

Table 5: Framework presence in C0 (bare prompt, 15 reviews) vs. C6 (full profile, 15 reviews). C0 counts are zero by construction: none of the six frameworks surface in unprompted reviews at any scoring level. C6 counts are drawn from the frequency analysis in the companion taxonomy paper [?].

Framework	C0 present	C6 present	Net gain
A-ha economy	0	8	+8
Load-bearing test	0	10	+10
World-model consistency	0	7	+7
Social fabric	0	4	+4
Perceptual-shift	0	9	+9
Visual grammar	0	6	+6

5 Discussion

5.1 The Symmetric Gradient

The authoring ablation produces the same non-monotonic gradient shape as the reviewer ablation in Paper I: more context does not monotonically improve quality, and the largest single inflection occurs not at the introduction of domain data but at the introduction of a design philosophy. The A0-to-A3 climb (+9.3 points across four conditions) mirrors the C0-to-C3 climb in Paper I almost exactly in proportional terms; the plateau-and-retreat at A4–A6 mirrors the C4–C6 flattening on the reviewer side.

The A2 result is the clearest structural parallel to Paper I's C2 finding. In the reviewer study, principles named without operational definitions produced critiques that were measurably worse-calibrated than richer contexts. Here, domain data without a design framework produces puzzles that score only 0.6 points above bare task specification (28.3 vs. 27.7) despite extensive world-data citations. In both cases, providing raw content — factual data in A2, principle names in C2 — without a conceptual framework for how to use it yields output less effective than a modest increase in structural guidance would produce. The mechanism is the same in both directions: domain expertise loaded into a prompt does not automatically become domain expertise in the output. The profile's framework determines what the model does with the information it holds.

The A4≈A5 result is an equally precise parallel to Paper I's C4≈C3 finding, which showed that a reviewer's worldview was evaluatively as rich as a full operational checklist. For authoring, a designer's generative starting position — where they begin, what question they ask first, what the finish condition looks like — captures the overwhelming majority of the quality value of a complete construction profile. The 1.3-point gap between A4 and A5 reflects a specific and narrow contribution: verification rigor. The full profile's construction log eliminates ambiguous item candidates and cross-checks competing answers, which protects execution quality within a chosen mechanism. It does not elevate the mechanism, alter the design strategy, or produce qualitatively different puzzles. For a designer with strong domain instincts, worldview is the rate-limiting resource; the checklist is insurance.

The symmetry between the two ablations has a broader implication. The mechanisms that determine profile quality — framework versus content, generative worldview versus procedural list — are not specific to the evaluation task. They appear to be structural properties of how profiles engage model behavior, applicable whenever a profile must guide a complex, open-ended generation task. The puzzle hunt domain provides unusually clean measurement conditions, but the gradient shape is likely general.

5.2 The Lens-Type Transfer Model

The three-designer comparison yields a *proposed* three-category taxonomy of how reviewing expertise transfers to authoring that is determined by lens type rather than by the designer's overall profile quality or depth. With one designer per category, this taxonomy cannot be empirically validated from the present data; it is offered as a theoretically motivated classification scheme requiring replication with a broader designer sample. This result is nonetheless theoretically significant: it means that the question "can this reviewer profile be repurposed for authoring?" has a principled answer based on the profile's internal structure, not its surface sophistication.

Operational lenses (Katz) are inherently generative. Each named test in Katz's reviewing vocabulary has a direct authoring-time analog that is neither a metaphor nor an approximation. "Would I want to solve this?" inverts to a constraint: design something you yourself would want to solve, and evaluate candidates by applying that test during construction. "Can the meta be short-circuited?" becomes: ensure no subset of four items resolves the answer without requiring the fifth. The Computer Test — eliminating clues a search engine can answer without domain knowledge — becomes a candidate filter applied before any clue is written. These inversions are lossless: the reviewing test and the authoring constraint are logically equivalent, distinguished only by the direction of application. The A6-Katz result (35.3/45, the WOLOLO puzzle) confirms this: the reviewer-as-author exercise produced the strongest puzzle among all Katz conditions precisely because the reviewing discipline sharpened item selection rather than being abandoned or approximated.

Construction lenses (Snyder) are already authoring-compatible. Snyder's reviewing questions — is each element load-bearing? does the solve path hold in exactly one direction? are there cross-clue dependencies that reward deductive rather than guessing strategy? — are indistinguishable from authoring instructions. The A5-to-A6 gap for Snyder is −0.7 points, within noise, because both profiles execute the same underlying checklist. For construction-oriented experts, the authoring/reviewing distinction is not a cognitive distinction; they apply the same tests in both directions. This collapses the practical difference between the A5 and A6 profiles for this lens type.

Perceptual lenses (Young) are terminal and format-dependent. Young's reviewing questions — does the visual hierarchy communicate structure before the content does? is the layout doing independent semantic work? does the visual grammar of the puzzle reinforce its domain? — require finished material in a format that can carry visual and experiential information. In a text-only acrostic, these checks have no application surface. The A5-Young puzzle (34.3/45) is strong but proceeds without Young's primary evaluative lens engaging at all; she substitutes phenomenological clue texture for her usual perceptual criteria. The A6-Young result (38.0/45, the highest score in the dataset) is therefore the finding that requires the most careful interpretation: it is not evidence that perceptual lenses transfer. It is evidence that blocking a designer's default approach — by forcing the reviewer-as-author exercise to surface the inapplicability of visual checks — can produce a productive substitution. Young found a format-compatible analog: second-person present-tense narrative that places the solver inside the AoE2 world rather than outside it. The perceptual instinct (make the solver feel inside the domain) found expression through narrative grammar because visual grammar was unavailable.

The practical implication for profile construction follows directly. When building authoring profiles from reviewing profiles: operational tests should be preserved and inverted; construction tests are transferable as-is; perceptual checks should be replaced by their generative precursors — not "does the layout communicate structure?" but "design the layout so its structure communicates before the content does." The last transformation requires identifying what the perceptual check is ultimately verifying (domain presence, experiential immersion, formal coherence) and constructing a generative instruction that can be satisfied during creation rather than only assessed after the fact.

A profile's lens type can be tentatively classified from its review lens section alone, without observing authoring behavior: count the proportion of questions framed as named operations with defined inputs and outputs (operational), as uniqueness or load-bearing tests (construction), and as perceptual or aesthetic assessments requiring finished material (perceptual). This procedure allows classification prior to any authoring study.

Two alternative explanations for the A5-to-A6 gap differences deserve acknowledgment. First, the gaps may reflect not lens type but profile length: Katz's C8 reviewing profile is longer than his authoring profile, which could independently account for A6-Katz outperforming A5-Katz. Second, the Young paradox (A6 > A5 despite an inapplicable lens) may reflect creative constraint effects specific to the text-acrostic format, and may not generalize to formats where her visual lens can be applied directly. Controlling for profile length and format compatibility are priorities for future work.

5.3 The Format Ceiling and the Riven Standard

All A-series puzzles in the authoring ablation score Riven Standard $\leq$ 3, regardless of profile depth or construction rigor. The first-letter acrostic mechanism is imported from puzzle hunt tradition and is not native to AoE2's game logic: the extraction step (take the first letter of each item name) has no analog in how AoE2 structures knowledge, rewards expertise, or produces meaning. The double-weighting of Elegance and Reading Reward in the /45 rubric penalizes format-imported mechanisms by design, since both dimensions reward domain-mechanism integration — the sense that the puzzle could not have been designed in any other domain. The Riven Standard rubric dimension makes this penalty explicit: a score of 3 indicates that the domain is content (the items are real AoE2 items) but not mechanism (the extraction logic is domain-agnostic).

The A6-Katz and A6-Young puzzles score Riven Standard 4, and A6-Young scores 5 — the only Riven 5 in the dataset. The distinction is mechanism adaptation: A6-Young constructs a coherent tactical scenario in which each item plays a role in a specific defensive situation, so the answer TOWER names not just a structure but the strategic position the solver has been defending across all five clues. This is the closest any evaluated puzzle comes to mechanism-domain integration without abandoning the acrostic format entirely.

The ceiling finding has a counterintuitive implication for profile depth. Profile depth raises the floor of quality within a fixed mechanism: the difference between A0 (21.7/45) and A5 (32.0/45) is real and large. But mechanism choice determines the ceiling, and no authoring profile in the A-series — not even the most sophisticated — can earn a high Riven Standard score when the extraction mechanism is format-imported. The companion persona study demonstrates this directly: a minimal two-word authoring persona — "you are an artist" — outperforms all A-series conditions (40.3/45, Riven Standard 4.3/5) by generating a mechanism native to the game's visual and archaeological register (damaged inscriptions with one missing letter per item, each gap corresponding to a domain-specific property). The artist persona did not produce a better-executed puzzle on any individual dimension; it produced a different class of mechanism. Profile depth and mechanism novelty are orthogonal contributions to quality, and the rubric is sensitive to both.

5.4 Implications for Computational Creativity

The symmetric gradient — authoring mirrors reviewing in both shape and inflection points — speaks directly to the *evaluation problem* in computational creativity [**?**]: if the same competence profile structure governs both generation quality and evaluation quality, then systems that evaluate well and systems that generate well may require the same underlying profile architecture, differing only in the direction of application. This has design implications: a reviewing profile is not a secondary artifact but a specification of the same latent competence that would govern authoring.

The A4≈A5 finding connects to Boden's account of *exploratory creativity* [**?**]: a designer who has internalized a generative worldview explores the same conceptual space as one equipped with a full operational checklist, because the worldview implicitly encodes the boundaries of that space. The checklist makes boundary conditions explicit; the worldview makes them operational. Either representation produces equivalent exploration of the design space when the designer is sufficiently expert.

The Young A6 result — where a perceptual lens blocked from its default application produced the highest-scoring output in the dataset — connects to the *creative constraint* literature [**?**]: constraints on a designer's natural approach can produce more creative outputs by blocking the default path and forcing a less-explored one. Young's perceptual instinct could not operate in the text-only format; the resulting substitution (narrative-grammar immersion for visual-grammar immersion) accessed a region of the design space her default approach would not have reached.

These findings position profile structure as a theoretically tractable variable in CC system design, not merely a prompt-engineering parameter. The companion persona-authoring-traits study takes this further by examining which profile features are predictive of mechanism novelty, the dimension most directly connected to Boden's transformational creativity account.

5.5 Limitations

Four limitations constrain the scope of these conclusions. First, all conditions use the same target answer (TOWER) and the same mechanism type (knowledge extraction via first-letter acrostic). The gradient findings may not generalize to other mechanism types — specifically to mechanisms where the extraction logic is already domain-native, which would lower the format ceiling for all conditions and potentially alter the relative contribution of profile depth. Second, the Young lens finding is format-specific. In a visually structured puzzle format — a grid, a map overlay, a layout-dependent extraction — Young's A6 profile would likely fail to achieve the forced substitution that produced the highest score in this dataset, and we predict it would underperform A5-Young in such formats. Third, the A5 and A6 authoring profiles used in the designer comparison were constructed for this study. They capture each designer's published design philosophy and stated principles but have not been validated against the designers' actual construction practice. The profiles are models of the designers, not the designers themselves. Fourth, evaluation calibration requires same-session scoring; all 13 puzzles in this study were read and ranked before any scores were assigned, following the calibration protocol established in Paper I. Scores from evaluation sessions conducted under different calibration conditions are not directly comparable to the values reported here, and earlier pilot scores for overlapping conditions have been superseded by the canonical c8-unified results.

A further limitation is that the three C8 panel reviewers (Katz, Snyder, Young) are also the designers whose profiles are used in Group B. This creates a potential self-evaluation confound: each reviewer evaluates puzzles partially authored under their own profile. We expect this produces conservative scores on A5 conditions (familiar constraints) and lenient scores on A6 conditions (recognizing their own lens), but the direction is uncertain.

6 Conclusion

This paper establishes three results about how expert persona profiles govern AI creative authoring. The first is the *symmetric profile effect*: the quality gradient produced by increasing authoring profile depth mirrors the non-monotonic reviewer ablation reported in Paper 1, with a salient inversion at the domain-expertise condition. Providing an AI author with domain world data alone (A2: 28.3/45) yields negligible improvement over the bare baseline (A0: 21.7/45) — a gap of 6.6 points driven almost entirely by the shift to an acrostic mechanism, not by domain content. Design principles (A3: 31.0) and a generative worldview (A4: 30.7) each add more value than the domain data does, and the worldview alone captures most of the value of the full construction profile (A5: 32.0), differing by only 1.3 points. The implication is consistent with Paper 1's reviewer finding: *framework* is the active ingredient in a profile; content without framework produces technically correct but experientially flat output. Why a design framework produces more quality lift than domain content remains an open question. One candidate explanation is that frameworks provide generative constraints that shape how content is selected and ordered, while raw content without framework produces the most salient associations rather than the most puzzle-appropriate ones — but this hypothesis requires a cognitive study to test directly.

The second result is the *lens-type transfer model*, which predicts whether a reviewing profile will improve, preserve, or degrade quality when repurposed for authoring. Operational lenses, exemplified by Katz's named testable checks (the Computer Test, the backwards read, the variety audit), transfer cleanly and improve output: Katz A6 outscores Katz A5 by 1.6 points, with WOLOLO: Five of a Kind (35.3/45) the strongest puzzle produced by any A-series designer. Construction lenses, exemplified by Snyder's load-bearing and uniqueness tests, are already generatively oriented; the A5-to-A6 transposition is essentially neutral (−0.7 points) because the reviewing and authoring profiles converge on the same questions. Perceptual lenses, exemplified by Young's world-model consistency and visual-grammar checks, are format-dependent: at A5 the acrostic format provides no purchase for her world-first instinct; at A6 the reviewer-as-author exercise forces a structural solution (second-person present tense, scenario-driven item selection) that the A5 profile could not reach, producing the highest single score in the dataset (Young A6: 38.0/45) and a gap of +3.7 points. The practical corollary: when adapting a reviewing profile for authoring, identify the lens type first — operational tests invert directly to authoring constraints; perceptual checks require reformulation as generative precursors before they transfer.

The third result concerns the *format ceiling*. Every A-series puzzle saturates at Riven Standard ≤ 3 because the first-letter acrostic mechanism was imported from puzzle-hunt tradition rather than derived from Age of Empires II. Profile depth raises the quality floor within a fixed mechanism — reducing arbitrary extraction, improving clue texture, tightening the solve path — but mechanism choice determines the ceiling. The Riven Standard dimension makes this boundary visible in a way the unweighted rubric did not; its double-weighting of Elegance and Reading Reward penalizes decorative domain content and rewards mechanism-domain integration.

The ceiling question — what authoring context produces mechanism novelty rather than incremental clue improvement — lies beyond the scope of this study but is addressed directly in the companion paper [**?**], where a bare artist persona outperforms all A-series conditions by inventing a domain-native mechanism rather than refining an imported one. Decomposing which traits of that persona drive the mechanism-generative behavior is the natural next step.

References